steady state voltage induced on no load is given by the intersection of magnetization characteristic of the machine and capacitance V-I characteristic.

PROCEDURE

Make the connections as shown in figure.

Precautions

i. Keep the dc machine (separately excited) field rheostat at maximum resistance position (Reason: Initially dc machine will act as dc generator)
ii. Keep DPST, SPST and TPST switches open
iii. Keep rotor rheostat at maximum resistance position

Switch on the three phase supply by closing the TPST switch. Induction machine will start as induction motor. As the motor gathers speed, gradually cut off the rotor resistance. Now, switch on the DC supply by closing the DPST switch. Decrease the resistance of the dc motor field rheostat (i.e., excitation is increased) and note the voltmeter reading across the SPST switch. If it is increasing, switch off the dc supply and interchange the armature terminals A & AA. Again switch on the dc supply keeping the motor field rheostat at maximum position. Decrease the resistance of the field rheostat gradually and make the voltmeter reading across the SPST switch zero. Now, the SPST switch is closed.

Again increase the excitation till the dc ammeter shows rated current. Please note that during this time, the connection of the reversing switch across the wattmeter should be such that the wattmeter reading is positive. Note down all the meter readings and time for 10 (or 5) oscillations. Now, decrease the excitation in steps for different values of dc ammeter and note all the readings each time. This procedure is continued till the wattmeter reads zero. Note the speed using tachometer.

If the excitation is again decreased, induction machine will be in generator mode. Now, interchange the connections of the pressure coil of the wattmeter using the reversing switch. Decrease the excitation and take readings for different values of dc ammeter. This procedure is continued till the dc ammeter reading reaches rated value (10A). Before switching off the dc supply, increase the excitation till the dc ammeter reads zero. Now switch off the ac supply also.